

		$I_M = \sqrt{\left(\frac{1}{2}\right)} I$
13	B	$PV = nRT$ At the bottom of the lake, $P(1.00 \times 10^{-6}) = nR (273.15 + 15) = nR(288.15)$ At the top water surface, $(1.00 \times 10^5)(5.00 \times 10^{-6}) = nR (273.15 + 30) = nR(303.15)$ $P = (1.00 \times 10^5)(5.00 \times 10^{-6}) (288.15) / (303.15) = 4.75 \times 10^5 \text{ Pa}$ Change in pressure = $(4.75 - 1.00) \times 10^5 = 3.75 \times 10^5 \text{ Pa} = 375 \text{ kPa}$
14	B	Higher specific heat capacity means that more energy will be needed to cool down or heat up